

Interaction Design in Digital Games

Pejman Mirza-Babaei
pejman.mirza-babaei@uoit.ca
Bluemoon UX Research and Development
Ontario, Canada

Samantha Stahlke
Samantha.Stahlke@gmail.com
Jam City
Ontario, Canada

ABSTRACT

This course covers user experience-focused approach to game interaction design. Created with the needs of aspiring game designers in mind, it's course suitable for HCI and games students, educators and researchers looking for a deeper understanding of how players interact with video games and how these interactions impact player experience. With hundreds of universities worldwide offering digital media and game development programs, interactive technology is a fast-growing field of study. This course fills a much-needed gap in the CHI's games and play community by connecting interaction design to fundamentals of user experience (UX) and general game design, addressing the needs of students and educators.

CCS CONCEPTS

• **Human-centered computing** → **Interaction design process and methods.**

KEYWORDS

Digital Games, Interaction Design, UX, UI, IxD

ACM Reference Format:

Pejman Mirza-Babaei and Samantha Stahlke. 2023. Interaction Design in Digital Games. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems (CHI EA '23)*, April 23–28, 2023, Hamburg, Germany. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3544549.3574178>

1 BENEFITS

Digital gaming has become a juggernaut in the landscape of modern entertainment. The global interactive software industry has grown rapidly in the past decade, with video game development forming a substantial part of this sector. Today, digital gaming is the largest entertainment industry in the world, growing at a rate of around 15 percent per year and reaching 137.9 billion USD in annual revenue worldwide. Beyond entertainment, products of the interactive media industry have spread to applications in healthcare, education, training, culture, art, and many other fields. Since many of these applications are dependent on business-to-consumer marketing (B2C), their success relies on whether users are able to have a fulfilling experience interacting with the product. This makes the subject of interaction design, the focus of this course, an essential and integral part of the development process. The massive growth

in the interactive entertainment industry has been accompanied by an equal acceleration in the popularity of educational programs offering related degrees. Students and educators in these programs form our primary audience for this course. It is important to note here that interaction design is critical for all types of programs, whether they are focused on programming, design, or art; there are often several courses on the design aspects of game and interactive app development. This course would contribute and enhance participants knowledge on topics related to game design, interaction design, user-centred design, and HCI.

2 INTENDED AUDIENCE(S)

This course aims to provide an introduction to interaction design and its applications in the digital entertainment industry. Type audience for the course is split into two components within the CHI community: 1) Students and educators in programs related to the design and development of digital products, interactive media, and video games; 2) Professionals (designers, developers and researchers) in interactive media industries, such as game and app development.

3 PREREQUISITES

N/A.

4 CONTENT

The course will be delivered in 2 units, and aims to provide a comprehensive analysis of game interaction design from an application perspective, suitable as both an introductory course for those new to the field while also covering a systematic approach to interaction design for more experienced participants. This balance is going to be achieved by first providing an essential introduction to interaction design, user-centered design, human-computer interaction for games covered in the first unit (an outline of this unit is provided below). In the second unit, we move on to present an in-depth analysis of different components of interaction design by exploring insightful design examples and case studies from industry researchers across several categories of game interaction. We explain several processes associated with successful game interaction design, providing a starting point for real-world applications. Topics covered in Unit 1:

- Intro to understanding players - different people play and interact in different ways, people have a wide variety of gameplay preferences, etc.
- How a better understanding of user interaction has helped to shape modern software design practices
- Why it is useful for games, briefly touch on differences between interaction in games versus productivity applications, how the case of games is different

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

CHI EA '23, April 23–28, 2023, Hamburg, Germany

© 2023 Copyright held by the owner/author(s).

ACM ISBN 978-1-4503-9422-2/23/04.

<https://doi.org/10.1145/3544549.3574178>

- What is interaction? (explanation of how humans engage with various objects/systems)
- Types of interaction (instructing, conversing, manipulating, exploring/browsing), different theories of interaction and how interactions can be analyzed in games, there are many types of interactions - can break down into intent, action, event, consequence.
- What makes a game interaction successful at a high level - understanding, ease of use, satisfying to perform, etc.

Topics covered in Unit 2:

- How to introduce players to mechanics, player learning (tutorials)
- How to reinforce concepts (building knowledge, remembering how to do things)
- How to keep players engaged/interested (skill curves, learning/improving, building regular play habits, achieving mastery)
- Importance of feedback - how output needs to clearly communicate the consequences of an action, player status, in-game events, etc.
- Iconography & consistency
- Affordances in the virtual world
- The process of designing input schemes - break down what actions are available, look at similarities across games/within a game, assign mappings, iterate and refine
- Intuitivity and player expectations
- Controller mappings and the issue of cross-platform controls
- Non-traditional input and novel hardware - how to design for new means of interaction

5 PRACTICAL WORK

We will engage participants in analysing the quality of interaction design by proving them examples from existing commercial games.

6 INSTRUCTOR BACKGROUND

The course is co-designed by **Pejman Mirza-Babaei** and **Samantha Stahlke**. They have been working together since 2015 on various commercial and academic research projects related to game and interaction design. They have co-authored many articles together and have a successful history of collaboration in work, research, and teaching. They co-authored a book on game interaction design (title: *The Game Designer's Playbook*) [1] which was published by Oxford University Press in August 2022. Their experiences and background (Samantha's deep understanding of game development and Pejman's extensive knowledge on human factors and interaction design) are complementary, making them an ideal team to deliver this course successfully.

Pejman Mirza-Babaei, PhD is an Associate Professor at Ontario Tech University and a Visiting Research Fellow at the University of Sussex (UK). His research and teaching is focused on the design and evaluation of user experience in interactive entertainment systems. He has published more than 50 peer-reviewed articles, and frequently speaks at academic conferences and industry events. Most of his projects are in collaboration with commercial game developers. Pejman has worked on pre- and post-release evaluation of more than 25 commercial games including award-winning titles

such as *Crysis 2*, *PewDiePie: Legend of the Brofist*, and *Weirdwood Manor*. He also served as UX Research Director at Execution Labs (Montréal, Canada) from 2015 - 2017 and was a UX researcher at Vertical Slice/Player Research (UK) from 2009 to 2013.

Samantha Stahlke is game artist at Jam City Inc. Toronto and Vertigo Gaming Inc. She has a Bachelor's degree (honours) in game development and experience in academic research, game programming, education, and commercial UX consultation. Her primary research interests include interaction design, human-computer interaction (HCI), games user research (GUR), artificial intelligence, and developing design tools. She has exhibited as an indie developer at several game conferences including PAX East and the Montreal International Games Summit. She frequently writes about game design and development, several of her writings have been featured on sites including Gamasutra and Game Career Guide.

7 RESOURCES

The course is based on the book written by the instructors: *The Game Designer's Playbook: An Introduction to Game Interaction Design*, Published by Oxford University Press, 2022 [1].

8 ACCESSIBILITY

To improve accessibility of the audience, we will provide all content and resources both in digital and printed format. We will test all digital content for accessibility to make sure they meet all requirements (e.g. readable by screen reader software). All video material will have subtitle.

REFERENCES

- [1] Samantha Stahlke and Pejman Mirza-Babaei. 2022. *The Game Designer's Playbook: An Introduction to Game Interaction Design*. OUP Oxford, Oxford.